Math 241, F18
Final Exam

Name: Answer

Instructions:

- There are totally 7 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	15	
2	15	
3	16	
4	14	
5	12	
6	14	
7	14	
Total	100	

Good Luck !

Problem 1. (15 points) Let $\mathbf{a} = 5\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} - \mathbf{k}$, and $\mathbf{c} =< 1, 6, 0 >$, Find each of the following:

(a) The cosine of the angle between $\mathbf{a}$ and $\mathbf{b}$.

$$|a| = \sqrt{5^2+(-2)^2+(5)^2} = \sqrt{54}$$

$$|b| = \sqrt{3^2+(0)^2+(-1)^2} = \sqrt{10}$$

$$a\cdot b = 5\cdot 3+(-2)\cdot 0+5\cdot(-1) = 10$$

$$\cos\theta = \frac{a\cdot b}{|a|\cdot|b|} = \frac{10}{\sqrt{54}\cdot\sqrt{10}} = \sqrt{\frac{10}{54}} = \sqrt{\frac{5}{27}} = \frac{\sqrt{5}}{3\sqrt{3}}$$

(b) $\mathbf{b} \times \mathbf{c}$.

$$b\times c = \begin{vmatrix} i & j & k \\ 3 & 0 & -1 \\ 1 & 6 & 0 \end{vmatrix} = i\begin{vmatrix} 0 & -1 \\ 6 & 0 \end{vmatrix} - j\begin{vmatrix} 3 & -1 \\ 1 & 0 \end{vmatrix} + k\begin{vmatrix} 3 & 0 \\ 1 & 6 \end{vmatrix}$$

$$= 6i + j + 18k$$

(c) The vector projection of $\mathbf{c}$ onto $\mathbf{a}$, $\text{Proj}_{\mathbf{a}}\mathbf{c}$.

$$a\cdot c = 5\cdot 1+(-2)\cdot 6+5\cdot 0 = -7$$

$$\text{Proj}_a c = \frac{c\cdot a}{|a|^2}a = \frac{-7}{54}\cdot <5,-2,5>$$

$$= <-\frac{35}{54}, \frac{7}{27}, -\frac{35}{54}>$$

Problem 2. (15 points)

(a) Find an equation of the plane through the point $(2, 5, 0)$ that is parallel to the plane $4x - 2y + z = 5$.

$$4(x-2) - 2(y-5) + (z-0) = 0$$
$$4x - 8 - 2y + 10 + z = 0$$
$$4x - 2y + z = -2$$

(b) Find an equation of the tangent plane to the surface $z = y(1 + \ln x)$ at the point $(1, 4, 4)$.

$$f(x,y,z) = z - y(1+\ln x) = 0$$
$$f_x = -\frac{y}{x}, \quad f_y = -(1+\ln x), \quad f_z = 1$$
$$\Rightarrow f_x(1,4,4) = -\frac{4}{1} = -4, \quad f_y(1,4,4) = -(1+\ln 1) = -1, \quad f_z(1,4,4) = 1$$
$$-4(x-1) - (y-4) + (z-4) = 0$$
$$-4x + 4 - y + 4 + z - 4 = 0$$
$$-4x - y + z = -4$$

(c) Let $f(x, y) = \sqrt{y + \cos^2 x}$, find an approximation of $f(0.05, 0.1)$ by using the linear approximation for f at (0,0).

$$f_x = \frac{1}{2}(y + \cos^2 x)^{-\frac{1}{2}}(2\cos x)(-\sin x) \Rightarrow f_x(0,0) = 0$$
$$f_y = \frac{1}{2}(y+\cos^2 x)^{-\frac{1}{2}} \cdot (1) = \frac{1}{2\sqrt{y+\cos^2 x}} \Rightarrow f_y(0,0) = \frac{1}{2}$$
$$\Rightarrow L(x,y) = f(0,0) + f_x(0,0)\cdot(x-0) + f_y(0,0)(y-0)$$
$$= \sqrt{0+1^2} + 0\cdot x + \frac{1}{2}y = 1 + \frac{1}{2}y$$

thus $f(0.05, 0.1) \approx L(0.05, 0.1) = 1 + \frac{1}{2}\cdot 0.1 = 1.05$

Problem 3. (16 points)

(a) Let $\mathbf{r}(t) = < e^t \cos t, e^t \sin t, e^t >$. Find the curvature κ of the curve.

$$r'(t) = < e^t \cos t - e^t \sin t, e^t \sin t + e^t \cos t, e^t >$$

$$= < e^t(\cos t - \sin t), e^t(\cos t + \sin t), e^t >$$

$$|r'(t)| = \sqrt{e^{2t}(\cos t - \sin t)^2 + e^{2t}(\cos t + \sin t)^2 + e^{2t}}$$

$$= e^t\sqrt{(1 - 2\cos t \sin t) + (1 + 2\cos t \sin t) + 1} = \sqrt{3}e^t$$

$$\Rightarrow T(t) = \frac{r'(t)}{|r'(t)|} = \frac{1}{\sqrt{3}} < \cos t - \sin t, \cos t + \sin t, 1 >$$

$$T'(t) = \frac{1}{\sqrt{3}} < -\sin t - \cos t, -\sin t + \cos t, 0 >$$

$$|T'(t)| = \frac{1}{\sqrt{3}}\sqrt{(-\sin t - \cos t)^2 + (-\sin t + \cos t)^2 + 0^2} = \frac{1}{\sqrt{3}}\sqrt{(1 + 2\cos t \sin t) + (1 - 2\cos t \sin t)}$$

$$= \frac{\sqrt{2}}{\sqrt{3}}$$

$$\Rightarrow k(t) = \frac{|T'(t)|}{|r'(t)|} = \frac{\frac{\sqrt{2}}{\sqrt{3}}}{\sqrt{3}e^t} = \frac{\sqrt{2}}{3e^t}$$

(b) Find the indicated limit or explain why it does not exist.

$$(1) \qquad \lim_{(x,y)\to(0,0)} \frac{xy}{x^2 + y^2}$$

Along the line $y = kx$ $(x \neq 0)$

$$\frac{xy}{x^2+y^2} = \frac{x(kx)}{x^2 + (kx)^2} = \frac{kx^2}{(1+k^2)x^2} = \frac{k}{1+k^2}$$

when $k = 1$, $\frac{k}{1+k^2} = \frac{1}{2}$

$k = -1$, $\frac{k}{1+k^2} = -\frac{1}{2}$

$\Rightarrow$ the limit doesn't exist.

$$(2) \qquad \lim_{(x,y)\to(0,0)} \frac{x^4 - y^4}{x^2 + y^2}$$

$$\lim_{(x,y)\to(0,0)} \frac{x^4 - y^4}{x^2+y^2} = \lim_{(x,y)\to(0,0)} \frac{(x^2+y^2)(x^2-y^2)}{x^2+y^2}$$

$$= \lim_{(x,y)\to(0,0)} x^2 - y^2 = 0$$

Problem 4. (14 points)
(a) Given a function $f(x,y) = 3xy^2 + 2\frac{x^2}{y^3}$. Find ∇f and the directional derivative of f at (2,1) in the direction of $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j}$.

$$\nabla f = \langle f_x, f_y \rangle = \langle 3y^2 + \frac{4x}{y^3}, 6xy - 6\frac{x^2}{y^4} \rangle$$

$$\Rightarrow \nabla f(2,1) = \langle 3\cdot 1^2 + \frac{4\cdot 2}{1^3}, 6\cdot 2\cdot 1 - 6\cdot\frac{2^2}{1^4} \rangle = \langle 11, -12 \rangle$$

$$u = \frac{a}{|a|} = \frac{\langle 3, -4 \rangle}{\sqrt{3^2+4^2}} = \langle \frac{3}{5}, -\frac{4}{5} \rangle$$

$$D_u f(2,1) = \nabla f(2,1)\cdot u = \langle 11, -12 \rangle \cdot \langle \frac{3}{5}, -\frac{4}{5} \rangle$$
$$= \frac{33}{5} + \frac{48}{5} = \frac{81}{5}$$

(b) Find all critical points of $f(x,y) = e^{-(x^2+y^2-4y)}$ and then determine whether each such a point gives a local maximum or minimum, or whether it is a saddle point.

$$\begin{cases} f_x = -2xe^{-(x^2+y^2-4y)} = 0 \\ f_y = -(2y-4)e^{-(x^2+y^2-4y)} = 0 \end{cases} \Rightarrow \begin{cases} -2x = 0 \\ -(2y-4) = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = 2 \end{cases}$$

Only one critical point $\underline{(0,2)}$.

$$f_{xx} = -2e^{-(x^2+y^2-4y)} + 4x^2e^{-(x^2+y^2-4y)} = (4x^2-2)e^{-(x^2+y^2-4y)}$$
$$f_{yy} = -2e^{-(x^2+y^2-4y)} + (2y-4)^2e^{-(x^2+y^2-4y)} = (4y^2-16y+14)e^{-(x^2+y^2-4y)}$$
$$f_{xy} = (-2x)(4-2y)e^{-(x^2+y^2-4y)}$$

$$\Rightarrow f_{xx}(0,2) = -2e^4, \quad f_{yy}(0,2) = -2e^4, \quad f_{xy}(0,2) = 0$$

$$\begin{cases} D(0,2) = f_{xx}(0,2)f_{yy}(0,2) - f_{xy}^2(0,2) = 4e^8 > 0. \\ f_{xx}(0,2) = -2e^4 < 0 \end{cases}$$

$\Rightarrow f(0,2)$ gives a local maximum

Problem 5. (12 points)

(a) Evaluate $\int_0^{\pi}\int_0^{\sin\theta} 2r\,dr\,d\theta$.

$$= \int_0^{\pi} \left[r^2\right]_{r=0}^{r=\sin\theta} d\theta$$

$$= \int_0^{\pi} \sin^2\theta\, d\theta = \int_0^{\pi} \frac{1-\cos 2\theta}{2}\, d\theta$$

$$= \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4}\right]_{\theta=0}^{\theta=\pi}$$

$$= \left[\frac{\pi}{2} - 0\right] - \left[0 - 0\right] = \frac{\pi}{2}$$

(b) Evaluate $\int_1^2\int_3^x\int_0^{\sqrt{3}y} \frac{2z}{y^2+z^2}\,dz\,dy\,dx$.

$$= \int_1^2\int_3^x \left[\ln(y^2+z^2)\right]_{z=0}^{z=\sqrt{3}y} dy\,dx$$

$$= \int_1^2\int_3^x \ln(4y^2) - \ln(y^2)\,dy\,dx$$

$$= \int_1^2\int_3^x (\ln 4)\,dy\,dx = (\ln 4)\int_1^2 \left[y\right]_{y=3}^{y=x} dx$$

$$= (\ln 4)\int_1^2 x-3\,dx$$

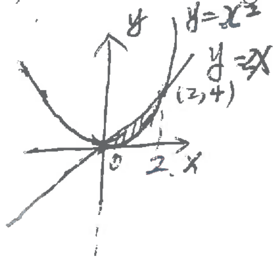
$$= (\ln 4)\left[\frac{x^2}{2} - 3x\right]_{x=1}^{x=2}$$

$$= -\frac{3}{2}\ln 4$$

$$= -\ln(4^{\frac{3}{2}}) = -\ln 8$$

Problem 6. (14 points)

(a) Let R be the region enclosed by $y = x^2$ and $y = 2x$, compute $\iint_R x+2y\, dA$.



$$\begin{cases} y=x^2 \\ y=2x \end{cases} \Rightarrow x^2=2x \Rightarrow x(x-2)=0 \Rightarrow x=0 \text{ or } x=2;\quad y=0 \quad y=4$$

$$\Rightarrow R=\{(x,y)\mid 0\le x\le 2,\ x^2\le y\le 2x\}$$

$$\iint_R 2x+y\,dA = \int_0^2\int_{x^2}^{2x} x+2y\,dy\,dx$$

$$= \int_0^2 [xy+y^2]_{y=x^2}^{y=2x}\,dx$$

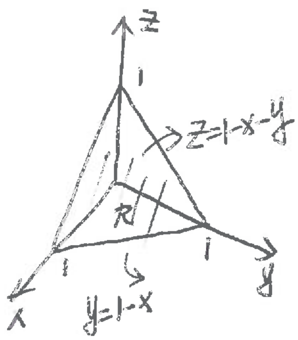
$$= \int_0^2 [(2x^2+4x^2)-(x^3+x^4)]\,dx$$

$$= \int_0^2 (6x^2-x^3-x^4)\,dx = \left[2x^3-\frac{x^4}{4}-\frac{x^5}{5}\right]_{x=0}^{x=2}$$

$$= 2\cdot 8-\frac{1}{4}\cdot 16-\frac{1}{5}\cdot 32$$

$$= 16-4-\frac{32}{5} = \frac{28}{5}$$

(b) Let T be the tetrahedron with vertices $(0,0,0), (1,0,0), (0,1,0)$, and $(0,0,1)$. Compute the volume of T by evaluating $\iiint_T dV$.



$$R=\{(x,y)\mid 0\le x\le 1,\ 0\le y\le 1-x\}$$

$$D=\{(x,y,z)\mid (x,y)\in R,\ 0\le z\le 1-x-y\}$$

$$\iiint_T dV = \int_0^1\int_0^{1-x}\int_0^{1-x-y} dz\,dy\,dx$$

$$= \int_0^1\int_0^{1-x} [z]_{z=0}^{z=1-x-y}\,dy\,dx = \int_0^1\int_0^{1-x} 1-x-y\,dy\,dx$$

$$= \int_0^1 \left[y-xy-\frac{y^2}{2}\right]_{y=0}^{y=1-x} dx$$

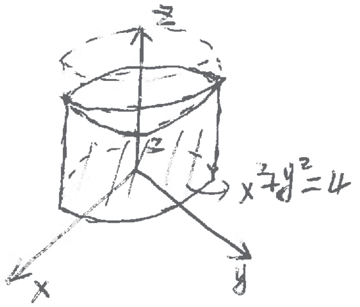
$$= \int_0^1 \left[(1-x)-x(1-x)-\frac{(1-x)^2}{2}\right]dx$$

$$= \int_0^1 \frac{1}{2}-x+\frac{x^2}{2}\,dx$$

$$= \left[\frac{1}{2}x-\frac{1}{2}x^2+\frac{1}{6}x^3\right]_{x=0}^{x=1}$$

$$= \frac{1}{2}-\frac{1}{2}-\frac{1}{6} = \frac{1}{6}$$

Problem 7. (14 points)
(a) Using the **cylindrical coordinates** to compute the triple integral $\iiint\limits_D \sqrt{x^2+y^2}\,dV$ where D is the solid that lies inside the cylinder $x^2+y^2=4$, and between the plane $z=0$ and the surface $z=2+x^2+y^2$.



$D=\{(r,\theta,z)\,|\,0\le r\le 2,\ 0\le\theta\le 2\pi,\ 0\le z\le 2+r^2\}$

$$\iiint\limits_D \sqrt{x^2+y^2}\,dV=\int_0^{2\pi}\int_0^{2}\int_0^{2+r^2} r\cdot r\,dz\,dr\,d\theta$$

$$=\int_0^{2\pi}\int_0^{2} r^2[z]_{z=0}^{z=2+r^2}\,dr\,d\theta$$

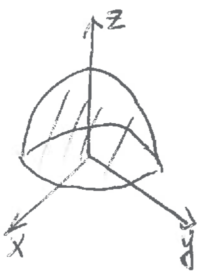
$$=\int_0^{2\pi}d\theta\int_0^2 r^2(2+r^2)\,dr$$

$$=2\pi\int_0^2(2r^2+r^4)\,dr$$

$$=2\pi\left[\frac{2}{3}r^3+\frac{1}{5}r^5\right]_{r=0}^{r=2}$$

$$=2\pi\left[\frac{2}{3}\cdot2^3+\frac{1}{5}\cdot2^5\right]=2\pi\left[\frac{16}{3}+\frac{32}{5}\right]=\frac{352}{15}\pi$$

(b) Let D be the hemisphere defined by $x^2+y^2+z^2\le 4$ and $z\ge 0$ with a density function $\delta(x,y,z)=z$. Find the mass of the solid D by evaluating $\iiint\limits_D z\,dV$ using the triple integral in the **spherical coordinates**.



$D=\{(\rho,\phi,\theta)\,|\,0\le\rho\le 2,\ 0\le\phi\le\frac{\pi}{2},\ 0\le\theta\le 2\pi\}$

$$\iiint\limits_D z\,dV=\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\int_0^2 \rho\cos\phi\,\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$$

$$=\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\int_0^2 \rho^3\cos\phi\sin\phi\,d\rho\,d\phi\,d\theta$$

$$=\int_0^{2\pi}\int_0^{\pi}\left[\frac{\rho^4}{4}\right]_{\rho=0}^{\rho=2}\cos\phi\sin\phi\,d\phi\,d\theta$$

$$=\int_0^{2\pi}\int_0^{\pi}4\cos\phi\sin\phi\,d\phi\,d\theta=\int_0^{2\pi}\int_0^{\pi}2\sin2\phi\,d\phi\,d\theta$$

$$=\int_0^{2\pi}d\theta\int_0^{\frac{\pi}{2}}2\sin2\phi\,d\phi=2\pi\cdot\left[-\cos2\phi\right]_{\phi=0}^{\phi=\frac{\pi}{2}}$$

$$=2\pi\cdot[1-(-1)]=4\pi$$